

GTU B.E. Computer Engineering - Artificial Intelligence Notes

Subject Code: 3170716 | Module 1 & 2: Fundamentals of AI & Problem Solving

1. Foundations of AI & Intelligent Agents

- Artificial Intelligence (AI) is the branch of computer science focused on creating systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, perception, and decision making.
- Intelligent Agent: An agent is anything that perceives its environment through sensors and acts upon it through actuators.
- PEAS Framework: Performance measure, Environment, Actuators, Sensors. Essential for defining agent tasks.
Example for Automated Taxi: P = Safety/Speed, E = Roads/Traffic, A = Steering/Brake, S = Cameras/Lidar.
- Types of Agents: Simple Reflex Agent, Model-based Reflex Agent, Goal-based Agent, and Utility-based Agent.

2. Search Algorithms in AI

- **Breadth-First Search (BFS):** Explores nodes level-by-level using a FIFO queue. It is complete.
Time Complexity: $O(b^d)$, Space Complexity: $O(b^d)$, where b is branching factor and d is depth.
- **Depth-First Search (DFS):** Explores deepest node first using a LIFO stack. Space complexity is $O(d)$.
- **A* Search:** Best-known form of heuristic search. Evaluation function: $f(n) = g(n) + h(n)$.
where $g(n)$ is the exact cost to reach node n , and $h(n)$ is the estimated heuristic cost from node n to the goal.
A* is complete and optimal if heuristic $h(n)$ is admissible (never overestimates true cost) and consistent.

GTU AI Notes - Module 3 & 4: Machine Learning & Classification

3. Supervised vs Unsupervised Learning

- **Supervised Learning:** Model is trained on labeled dataset where both inputs (X) and ground truth (Y) are known.
Common tasks: Classification (Spam detection, Disease diagnosis) and Regression (House price prediction)
- **Unsupervised Learning:** Finds hidden patterns, structures, and groupings in unlabeled data.
- **Bias-Variance Tradeoff:** High Bias leads to Underfitting (oversimplified model). High Variance leads to Overfitting (overly complex model).
- **Overfitting Prevention Techniques:** Regularization (L1 Lasso, L2 Ridge), Cross-Validation

4. Decision Trees & Entropy

- **Decision Tree:** Non-parametric supervised learning method used for classification and regression.
- **Entropy:** Measures impurity/uncertainty in a dataset. Formula: $H(S) = - \sum (p_i * \log_2(p_i))$
- **Information Gain (IG):** Expected reduction in entropy from partitioning data on an attribute.
- The attribute with the highest Information Gain is chosen as the splitting root node at each step.